

Sentiment Analysis Evaluation Report

SOTA Transformer Model vs. Linear SVC (Bag-of-Words)

Automotive Social Media Comments Dataset

April 2026

1. Introduction

This report documents the evaluation and comparison of two sentiment analysis systems applied to a dataset of automotive-related social media comments. The central question is whether a state-of-the-art (SOTA) pre-trained transformer model can outperform a classical machine learning baseline — specifically, a Linear Support Vector Classifier (SVC) trained on Bag-of-Words (BoW) features — on a domain-specific sentiment classification task.

Sentiment analysis on automotive social media content presents unique challenges. Comments often reference specific car models, technical topics such as battery range or charging infrastructure, and employ informal language, slang, and comparative expressions. Accurate classification into negative, neutral, and positive categories is valuable for understanding consumer perception, product feedback, and brand reputation in the electric and conventional vehicle market.

The SOTA model used in this evaluation is `cardiffnlp/twitter-roberta-base-sentiment-latest`, a RoBERTa-based model fine-tuned on Twitter data for sentiment analysis. It is loaded locally via the Hugging Face Transformers pipeline. The baseline is a Linear SVC model trained on a custom Bag-of-Words vocabulary derived from the same automotive comments dataset, loaded via `scikit-learn`.

The evaluation is structured around three goals:

1. Fair head-to-head comparison on a held-out test set (20% of the dataset).
2. Granular analysis of model behaviour across conversation topics (battery, range, price, etc.).
3. Granular analysis across car models (BMW, Tesla, Ford, etc.).

2. Methodology

2.1 Dataset

The dataset consists of cleaned social media comments stored in `cleaned_comments3.csv`. Each comment has been pre-processed into a `clean_comment` field and carries a `majority_vote_final` label derived from human annotators. Labels were normalised to lowercase and a known typo — 'negtaive' — was corrected to 'negative'. The three target classes are: negative, neutral, and positive.

2.2 Train / Test Split

A stratified 80/20 train-test split was applied using a fixed random seed (`random_state=42`) to ensure reproducibility and class-balance preservation across both partitions. The test set contains 201 samples and is the sole basis for the head-to-head model comparison. Note: the SVC model was trained on the training portion of this split; the SOTA model is a zero-shot application (it has not seen any training data from this dataset).

2.3 SOTA Model — RoBERTa (Transformer)

The SOTA model is loaded from a local directory (`sota_model/`) using the Hugging Face transformers pipeline API with the text-classification task. Inference is run on CPU (`device=-1`). The model outputs a label that maps directly to the dataset labels without additional post-processing — no label remapping was required.

2.4 Baseline Model — Linear SVC (Bag-of-Words)

The baseline is a scikit-learn Linear SVC model, serialised and loaded from `BEST_MODEL_F1_trial.pkl`. Input text is vectorised using a `CountVectorizer` initialised with a fixed vocabulary loaded from `scheme3_vocab.txt` — ensuring that the vocabulary is identical to what was used during training. Predictions are made on the vectorised test features.

2.5 Evaluation Metrics

All comparisons are performed using macro-averaged F1 score as the primary metric, ensuring equal weight across all three sentiment classes regardless of class imbalance. Accuracy and per-class precision, recall, and F1 are reported as supplementary metrics. Confusion matrices are generated for both models. Weighted F1 is also computed for reference but explicitly noted as a more favourable (less rigorous) metric when class distributions are skewed.

2.6 Subgroup Analysis

To understand model behaviour beyond aggregate scores, two metadata features are extracted via regex from each comment:

- Topic — one of: battery, range, price, charging, autopilot, design, performance, reliability, fuel. Comments not matching any keyword are labelled 'Other'.
- Car Model — matched against a curated list of 22 brands and models (e.g., Tesla, BMW, Ford, Hyundai). Unmatched comments are labelled 'Other'.

Both models are then evaluated independently on each topic and car-model subgroup, and the results are compared side-by-side. It is explicitly acknowledged in the notebook that the SVC's subgroup scores on the full dataset are inflated because the model has already been trained on most of those samples — the test-set comparison is therefore the definitive benchmark.

3. Quantitative Results

3.1 Head-to-Head Comparison on the Test Set (n = 201)

The table below summarises the primary evaluation results on the held-out test set:

Metric	SOTA (RoBERTa)	Linear SVC
Macro F1	0.645	0.671
Accuracy	68.2%	74.6%
Weighted F1	0.683	0.731
Neg. Precision / Recall / F1	0.48 / 0.57 / 0.52	0.65 / 0.35 / 0.46
Neu. Precision / Recall / F1	0.72 / 0.79 / 0.75	0.75 / 0.92 / 0.83
Pos. Precision / Recall / F1	0.80 / 0.57 / 0.67	0.78 / 0.69 / 0.73

The Linear SVC achieves a higher macro F1 (0.671 vs 0.645) and higher accuracy (74.6% vs 68.2%) on the test set. However, both models exhibit notable class-specific trade-offs. The SOTA model demonstrates stronger recall on negative comments (0.57 vs 0.35), making it less prone to missing negative sentiment. The SVC dominates on the neutral class, achieving 0.92 recall, which largely drives its accuracy advantage given that neutral is the majority class (103 of 201 test samples).

3.2 Confusion Matrices

The confusion matrices for both models on the test set (rows = true label, columns = predicted label, order: negative / neutral / positive) are:

Model	True Neg → Pred	True Neu → Pred	True Pos → Pred
SOTA (RoBERTa)	21 neg 16 neu 0 pos	13 neg 81 neu 9 pos	10 neg 16 neu 35 pos
Linear SVC	13 neg 17 neu 7 pos	3 neg 95 neu 5 pos	4 neg 15 neu 42 pos

The SOTA model misclassifies more neutral comments as negative (13) but fewer as positive (9 vs 5). The SVC model is strongly biased toward neutral predictions, consistent with its very high neutral recall. The SOTA model distributes errors more evenly across classes.

3.3 Topic-Level Subgroup Performance (Full Dataset)

When evaluated per topic across the full dataset, the SVC generally scores higher — a result that is expected given training data overlap. However, a notable finding is that the SOTA model approaches or matches the SVC on the battery, fuel, and performance topics, suggesting the transformer's pre-trained representations capture domain-relevant nuance in these subgroups. Price and range topics are the most populated (after 'Other') and show both models achieving moderate performance.

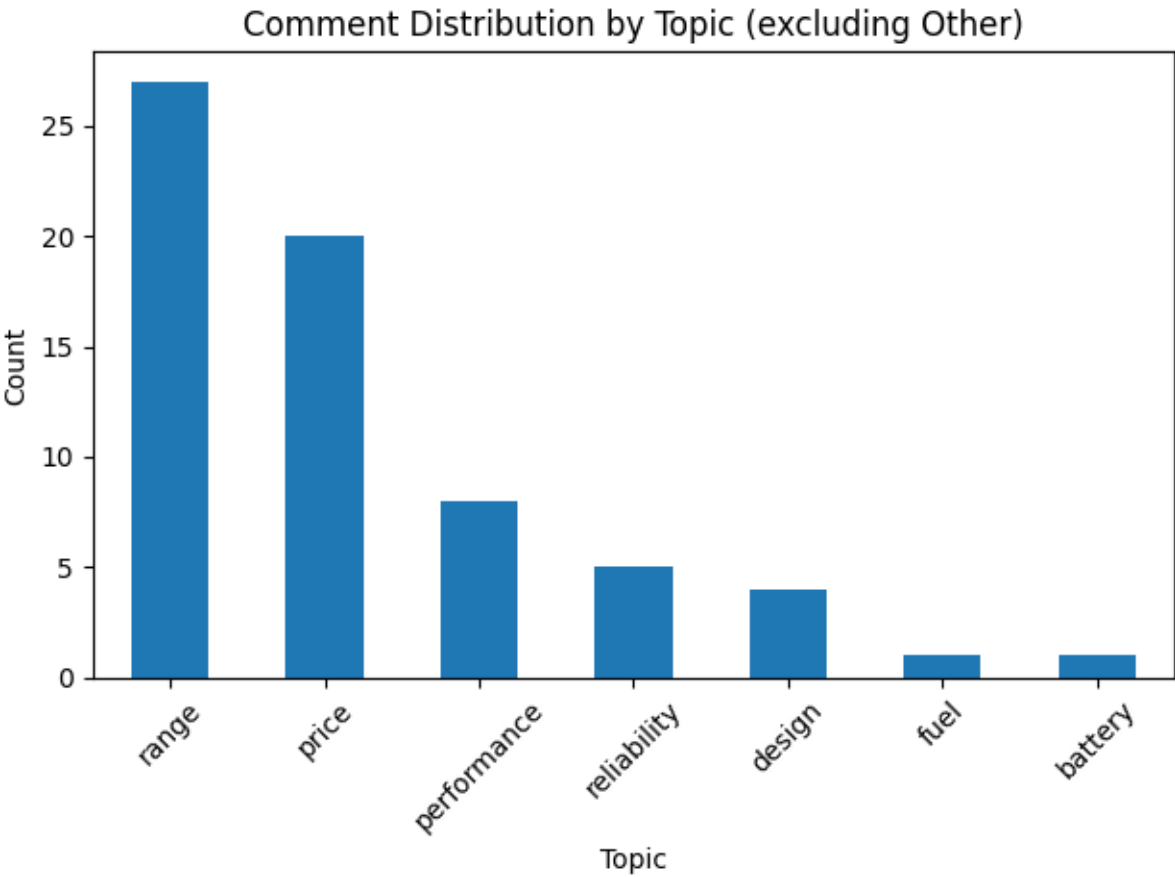
3.4 Car-Model Subgroup Performance

BMW is the most frequently mentioned brand in the dataset by a large margin. Performance across car models is variable for both models, largely due to small sample sizes in most subgroups. Results for low-frequency brands should be interpreted with caution.

4. Visual Analysis

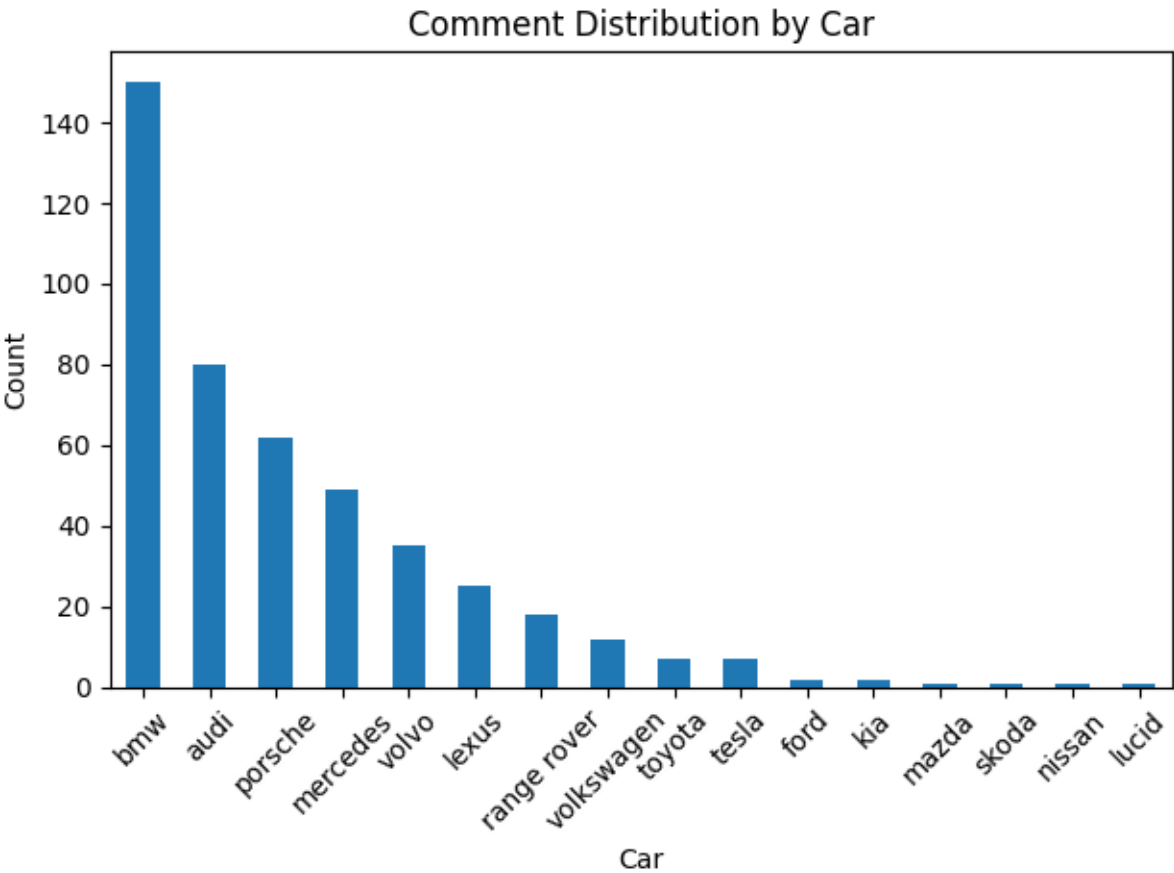
4.1 Comment Distribution by Topic

A bar chart of topic frequencies reveals that the 'Other' category accounts for the vast majority of comments — i.e., most comments do not mention any of the nine tracked keywords explicitly. Among the named topics, price and range are the most discussed, followed by battery and performance. This indicates that consumers discussing automotive products are primarily concerned with economic value and driving range, which aligns with known concerns in the electric vehicle market.



4.2 Comment Distribution by Car Model

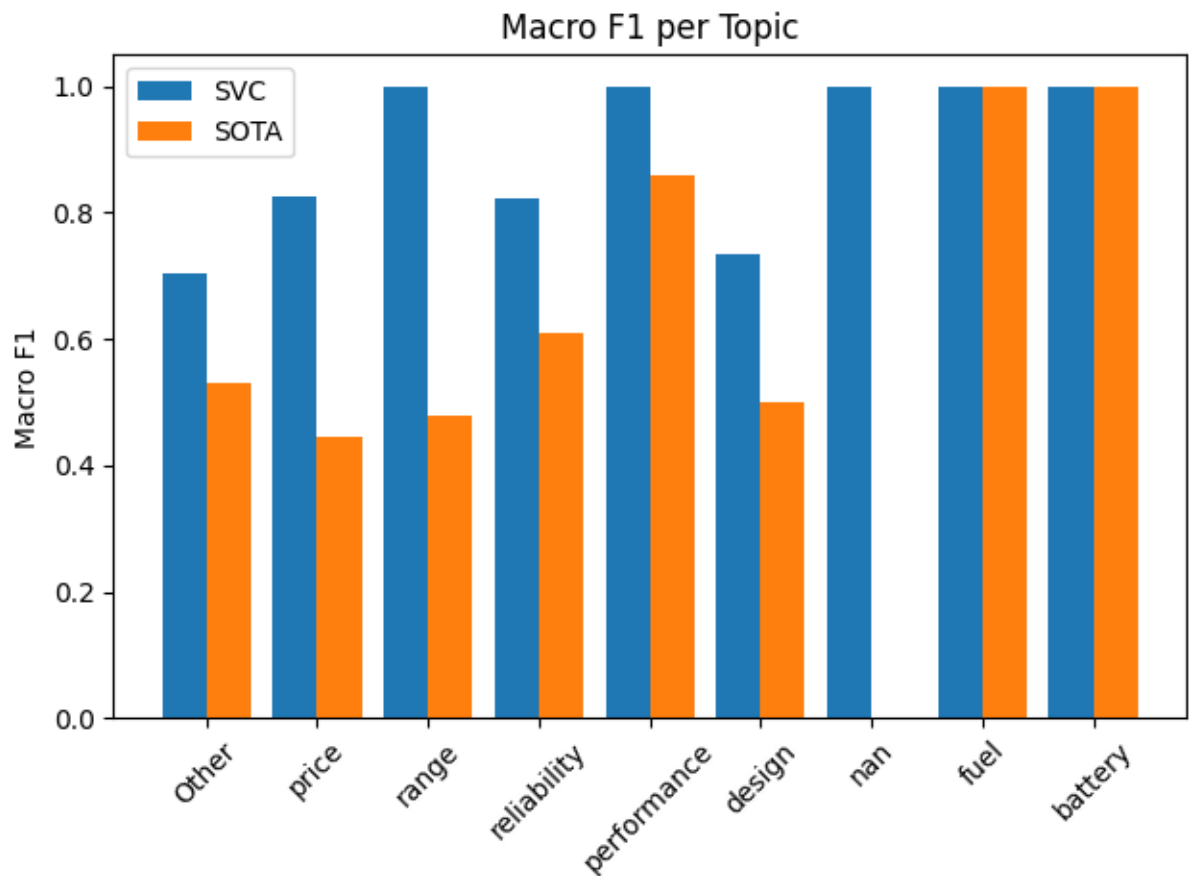
The car model distribution chart shows a heavily skewed distribution. BMW dominates comment volume among named brands, with Tesla, Ford, and a handful of others following at much lower frequencies. This skew has implications for subgroup analysis reliability: inferences about less-mentioned brands (e.g., Lucid, Rivian, Renault) are based on very few samples and may not be statistically meaningful.



4.3 Macro F1 per Topic — Side-by-Side Bar Chart

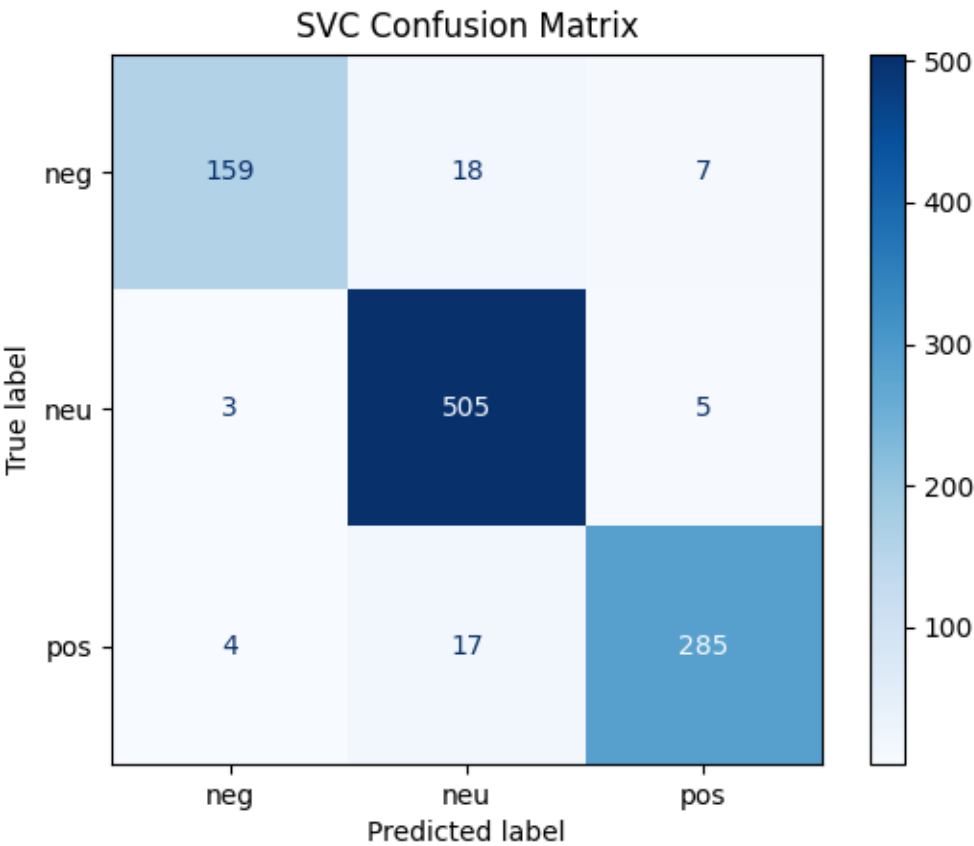
A grouped bar chart compares macro F1 per topic for SVC (blue) and SOTA (orange). Key observations:

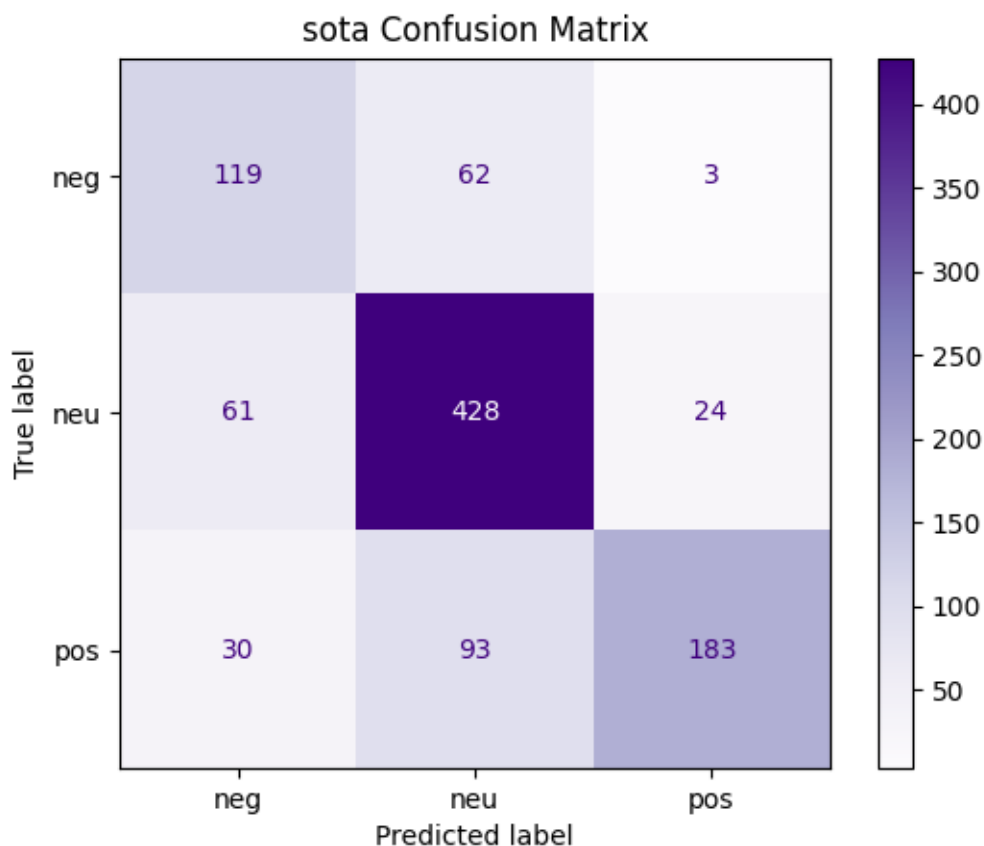
- SVC leads on most topics, but this advantage is confounded by training-set exposure.
- SOTA is competitive or superior on battery, fuel, and performance — domains where semantic understanding of technical language matters.
- Both models struggle on small-sample topics, reflecting data sparsity rather than model limitations.



4.4 Confusion Matrices — Heatmap Visualisation

Confusion matrix heatmaps (SVC in blue, SOTA in purple) confirm the quantitative findings. The SVC matrix shows a strong diagonal concentration in the neutral cell, reflecting the model's tendency to over-predict neutral. The SOTA matrix shows a more balanced but noisier pattern, with non-trivial off-diagonal values for negative comments — the model is more willing to predict negative but also makes more mistakes in that direction.

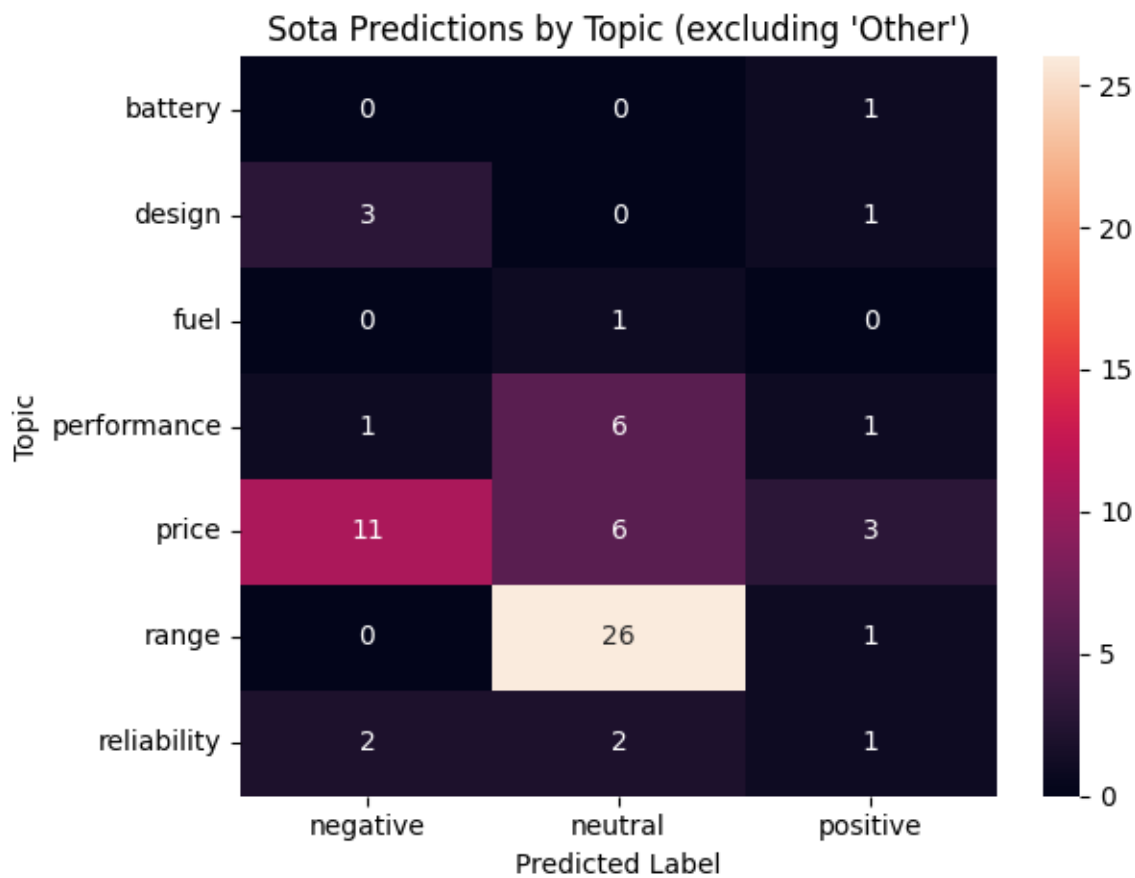


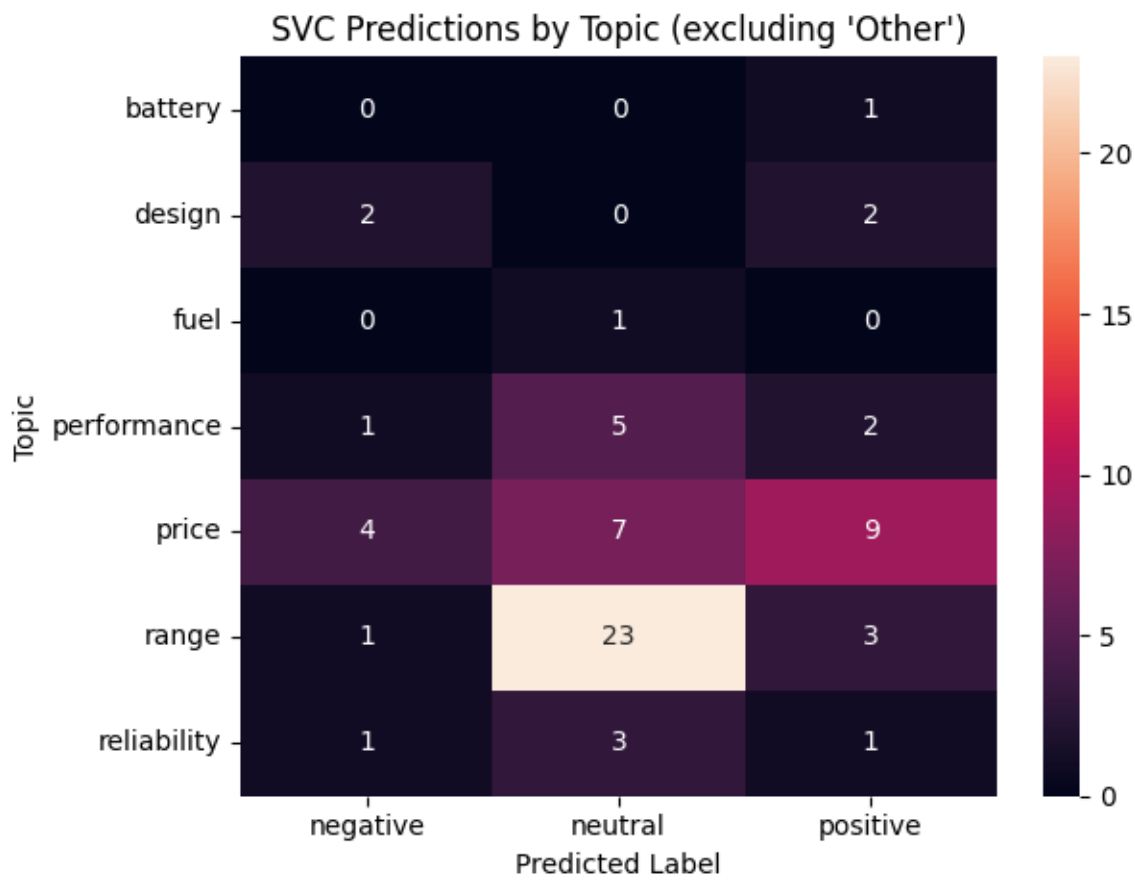


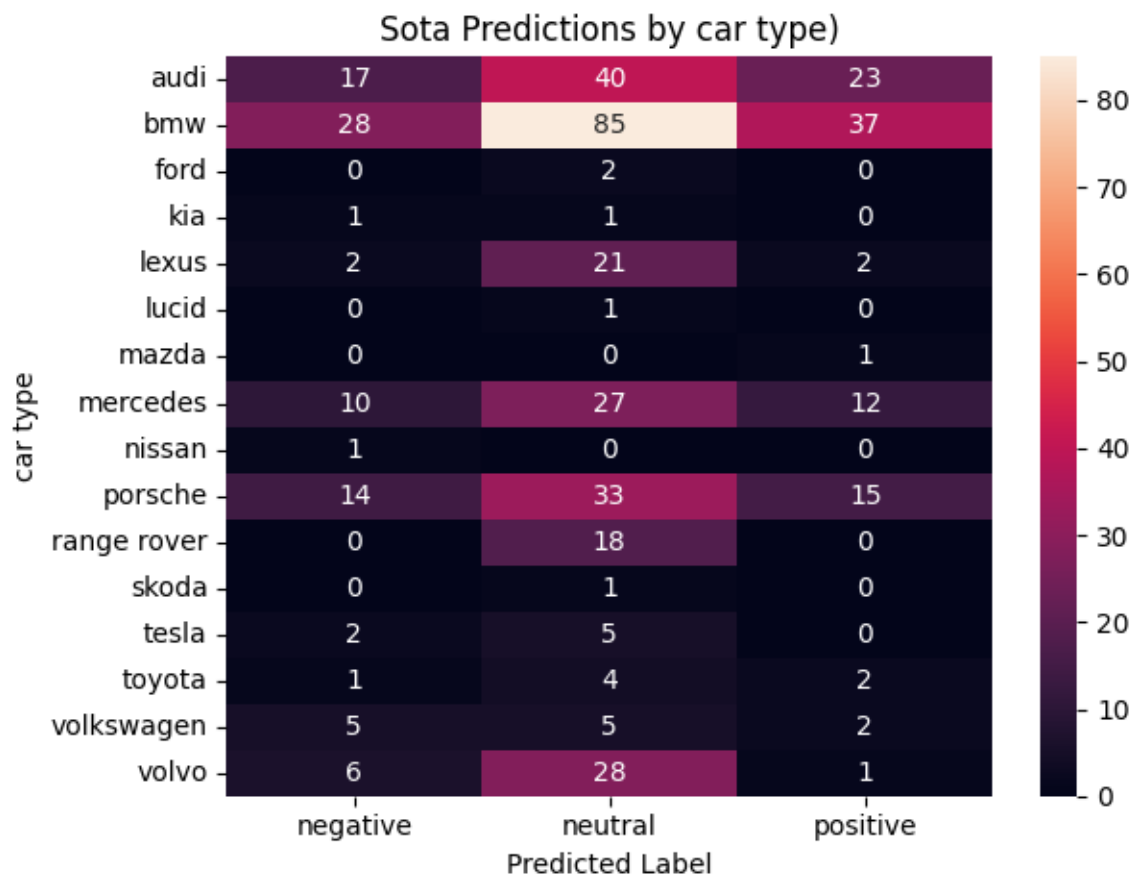
4.5 Prediction Heatmaps by Topic and Car Model

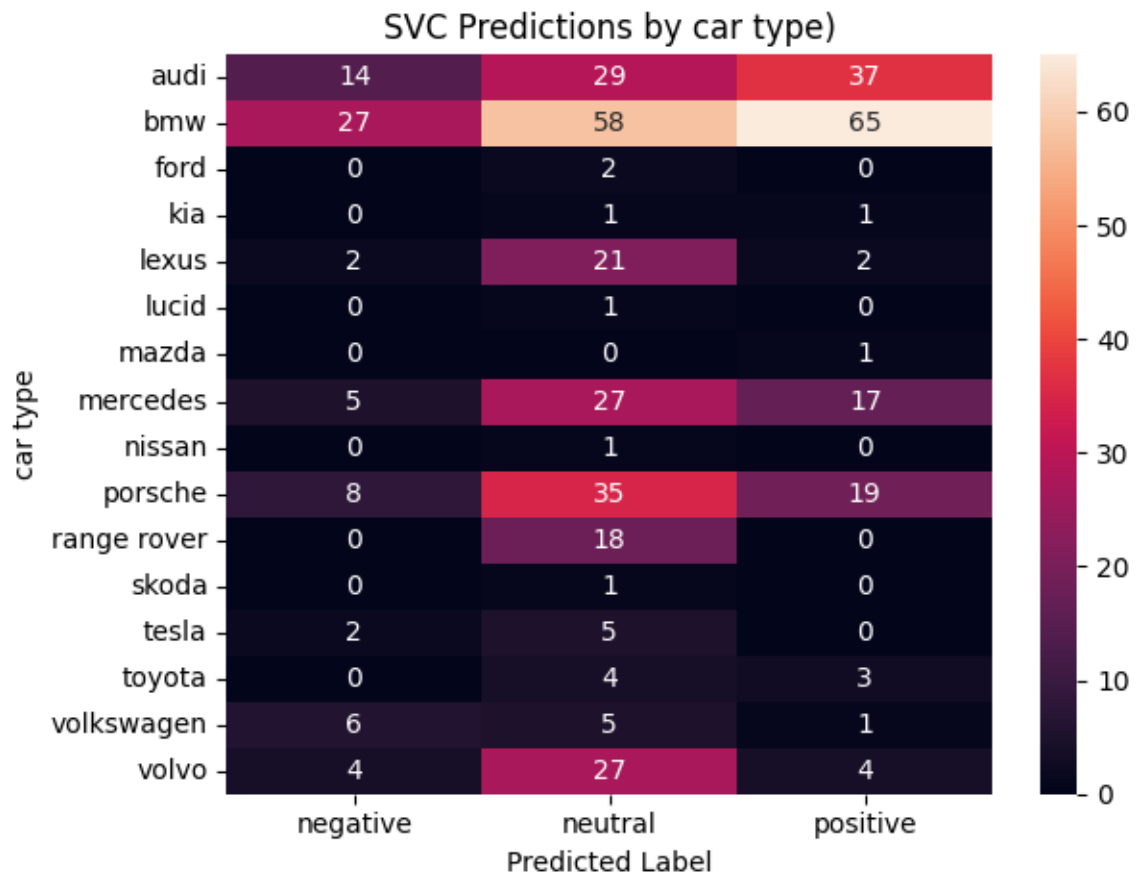
Seaborn heatmaps cross-tabulate topic (or car model) against predicted sentiment label for each model. Key findings:

- In both models, the neutral class is predicted far more frequently than negative or positive across virtually all topics, indicating a class imbalance effect that neither model fully overcomes.
- Price-related comments have the highest absolute prediction counts (after Other), with neutral being the dominant predicted label — suggesting price discussions tend toward factual, non-emotional language.
- For car models, BMW comments are most frequently predicted as neutral or positive across both models, consistent with BMW's generally positive brand perception in the dataset.
- The SOTA model shows slightly more varied distribution across positive/negative classes compared to the SVC, which concentrates heavily on neutral.









5. Conclusion

This evaluation compared a state-of-the-art pre-trained transformer (RoBERTa, fine-tuned on Twitter sentiment) against a domain-trained Linear SVC with Bag-of-Words features on the task of sentiment classification for automotive social media comments.

On the held-out test set — the only fair comparison point — the Linear SVC achieves a higher macro F1 (0.671 vs 0.645) and accuracy (74.6% vs 68.2%). This result, while perhaps counterintuitive given the general dominance of transformer architectures, is well-explained by the experimental context: the SVC was trained on in-domain data using a carefully constructed vocabulary, while the SOTA model is applied zero-shot to a domain (automotive social media) that differs from its Twitter fine-tuning data. The SVC's advantage is particularly pronounced on the neutral class, where its recall of 0.92 far exceeds the SOTA's 0.79.

Despite its lower aggregate scores, the SOTA model shows competitive and in some cases superior performance on technically-nuanced topics such as battery, fuel, and performance. This suggests that the transformer's pre-trained semantic representations do capture domain-

relevant information, and that with fine-tuning on labelled automotive data, the SOTA model could be expected to surpass the SVC.

Both models are affected by the class imbalance in the dataset — neutral comments constitute more than half the test set — resulting in a tendency to over-predict neutral. This is a structural challenge that would likely require oversampling, class weighting, or data augmentation strategies to address.

In summary, the key takeaways from this evaluation are:

- The Linear SVC, when trained on representative in-domain data, is a strong baseline that a zero-shot SOTA transformer does not reliably surpass.
- The SOTA model's stronger negative-class recall makes it more sensitive to critical feedback, which may be desirable in certain application contexts.
- Fine-tuning the SOTA model on this dataset is the most promising path to improving performance beyond what the SVC achieves.
- Price and range are the dominant conversation topics, and BMW is the most-discussed brand — findings with direct relevance to product and marketing strategy.

Future work should explore fine-tuning the transformer model on annotated automotive comments, addressing class imbalance through data augmentation, and expanding the keyword lists for topic and brand extraction to improve subgroup coverage.